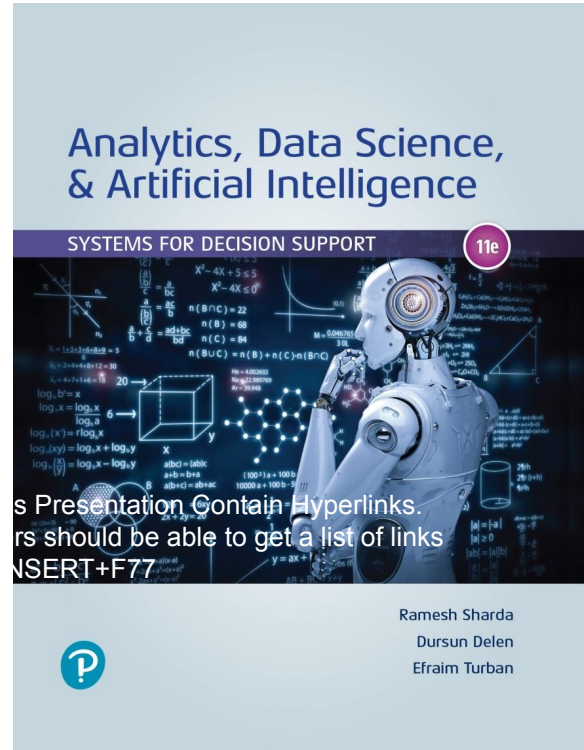


Analytics, Data Science and AI: Systems for Decision Support

Eleventh Edition

Chapter 9

Big Data, Cloud Computing, and
Location Analytics: Concepts and
Tools



s Presentation Contain Hyperlinks.
rs should be able to get a list of links
INSERT+F77

Learning Objectives (1 of 2)

- 9.1 Learn what Big Data is and how it is changing the world of analytics
- 9.2 Understand the motivation for and business drivers of Big Data analytics
- 9.3 Become familiar with the wide range of enabling technologies for Big Data analytics
- 9.4 Learn about Hadoop, MapReduce, and NoS Q L as they relate to Big Data analytics
- 9.5 Compare and contrast the complementary uses of data

Learning Objectives (2 of 2)

- 9.6** Become familiar with in-memory analytics and Spark applications
- 9.7** Become familiar with select Big Data platforms and services
- 9.8** Understand the need for and appreciate the capabilities of stream analytics
- 9.9** Learn about the applications of stream analytics
- 9.10** Describe the current and future use of cloud computing in business analytics

Why ?

glassdoor

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Awards

Best Places to Work

Highest Rated CEOs

Lists

Best Jobs

Best Cities for Jobs

Oddball Interview Questions

Trends

Overview

Job Trends

Trends by Industry

Trends by Location

Company Trends

Trends by Industry

This report ranks jobs according to each job's Glassdoor Job Score, determined by combining three factors: number of job openings, salary, and overall job satisfaction rating.

Employers: Want to recruit better in 2017? [Find out how.](#)

United States

2017

11k JOBS



1

Data Scientist



4.8 / 5

Job Score

4.4 / 5

Job Satisfaction

\$110,000

Median Base Salary

4,184

Job Openings

View Jobs

2

DevOps Engineer



Data Size

Name	Symb ol	Value	
Kilobyte	k B	10^3	1,000
Megabyte	M B	10^6	1,000,000
Gigabyte	G B	10^9	1,000,000,000
Terabyte	T B	10^{12}	1,000,000,000,000
Petabyte	P B	10^{15}	1,000,000,000,000,000
Exabyte	E B	10^{18}	1,000,000,000,000,000,000
Zettabyte	Z B	10^{21}	-
Yottabyte	Y B	10^{24}	-
Brontobyte*	B B	10^{27}	-
Gegobyte*	G e B	10^{30}	-

Big Data - Definition and Concepts

- Big Data is a misnomer!
- Big Data is more than just “big”
- The Vs that define Big Data
 - **Volume**: large amounts of data
 - **Variety**: data comes in many different forms from diverse sources
 - **Velocity**: the content is changing quickly
 - **Value**: data alone is not enough; how can value be derived from it?
 - **Veracity**: can we trust the data? how accurate is it?

Variety

well-defined fields

Structured Data

SUMMER FOOD SERVICE PROGRAM 1)				
(Data as of August 01, 2011)				
Fiscal Year	Number of Sites	Peak (July) Participation	Meals Served	Total Federal Expenditures 2)
	-----Thousands-----		--Mil--	--Million \$--
1969	1.2	99	2.2	0.3
1970	1.9	227	6.2	1.8
1971	3.2	569	29.0	8.2
1972	6.5	1,080	73.5	21.9
1973	11.2	1,437	65.4	26.6
1974	10.6	1,403	63.6	33.6
1975	12.0	1,786	84.3	50.3
1976	16.0	2,453	104.8	73.4
TQ 3)	22.4	3,455	198.0	88.9
1977	23.7	2,791	170.4	114.4
1978	22.4	2,333	120.3	100.3
1979	23.0	2,126	121.8	108.6
1980	21.6	1,922	108.2	110.1

by humans for humans

Unstructured Data

The Red Wheelbarrow, by William Carlos Williams

so much depends upon

a red wheel

barrow

glazed with rain

water

beside the white

chickens.



Semi-Structured Data



View → Source



```

1 1
2 <!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-trans
3 <html xmlns="http://www.w3.org/1999/xhtml">
4
5 <head>
6 <meta http-equiv="Content-Type" content="text/html; charset=UTF-8" />
7 <META NAME="y_key" content="55984020e1c9ac0c">
8 <link rel="canonical" href="http://www.emc.com/index.htm" />
9 <META NAME="description" content="EMC is a leading provider of storage hardware solutions th
10 <title>EMC - Data Recovery, Cloud Computing, and Storage Hardware</title>
11 <META NAME="description" content="EMC is a leading provider of storage hardware solutions th
12 Data recovery and improve cloud computing." />
13 <META NAME="keywords" content="emc, network storage, data recovery, information manage
14 software, nas storage, information protection, information management" />
15 <!-- Start stylesheet includes -->
16
17 1-CMPS653-S21-Session1-Welcome to Big Data!.pdf 1 of 9

```

Getting bigger

THE INTERNET IN **2023** EVERY MINUTE



Velocity type

Velocity: the content is changing quickly

- **Batch processing:** running a series of computer programs (in the background) without human intervention.
- **Near real-time:** brief delay between the data becoming available and it being processed.
- **Real-time:** immediate processing of data once available.

Example of volume and velocity

- **>500 million** tweets a day
- **>300 million** active users / day
- On average **>6000** tweets a second
- Peaks of **>100,000** tweets a second
 - Super Bowl
 - US election (**75M** tweets in 24h)
 - New Year's Eve
 - Football World Cup (**672M** tweets in total #WorldCup)
- Messages are instantly accessible for **search**
- Messages are used in **post-hoc** analyses to gather insights



Example of variety and velocity

- **Task:** Given a person's location anywhere in the world, list the top five restaurants in his immediate neighbourhood.
- **Required data:**
 - World map, list of all restaurants in the world (opening hours, GPS coordinates, menu, special offers)
 - Reviews/ratings
 - Optional: social media stream(s)
- Data is continuously **changing** (restaurants close, new ones open, data formats change, etc.)

Where Does Big Data Come From?

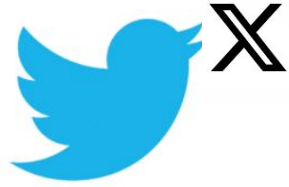


Online recording

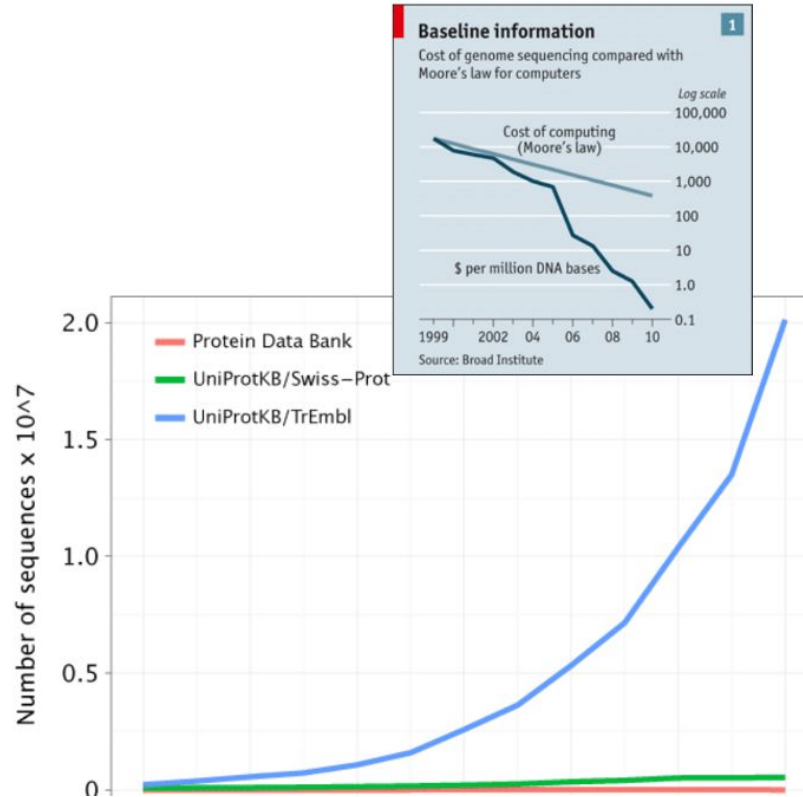
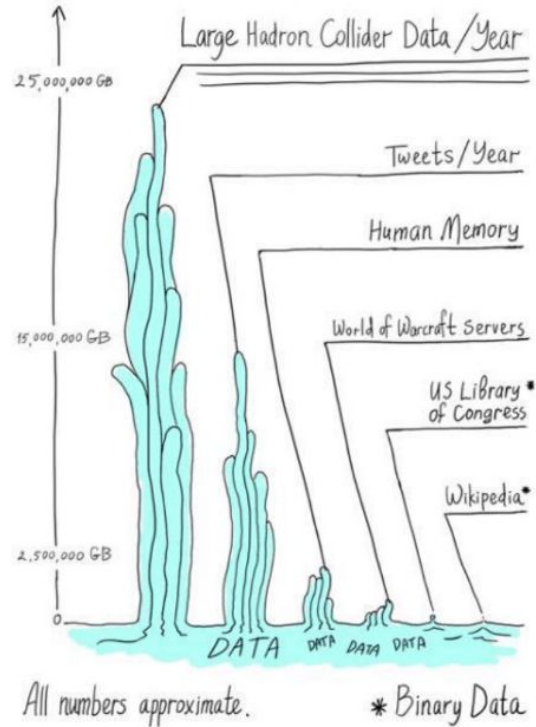
- It's all happening online – could record every:
 - Click
 - Ad impression
 - Billing event
 - Fast Forward, pause,...
 - Server request
 - Transaction
 - Network message
 - Fault
 - ...



User Generated



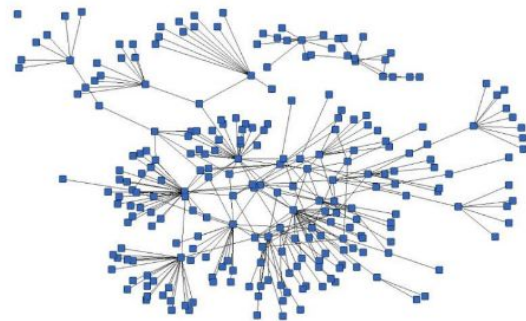
Health and science computations



Graph Data

Lots of interesting data has a graph structure

- Social networks
- Telecommunication Networks
- Computer Networks
- Road networks
- Collaborations/Relationships
- ...



Some of these graphs can get quite large! (e.g., Facebook user graph)

Log file

```
uplherc.upl.com - - [01/Aug/1995:00:00:07 -0400] "GET / HTTP/1.0" 304 0
uplherc.upl.com - - [01/Aug/1995:00:00:08 -0400] "GET /images/ksclogo-medium.gif
HTTP/1.0" 304 0
uplherc.upl.com - - [01/Aug/1995:00:00:08 -0400] "GET /images/MOSAIC-logosmall.gif
HTTP/1.0" 304 0
uplherc.upl.com - - [01/Aug/1995:00:00:08 -0400] "GET /images/USA-logosmall.gif HTTP/
1.0" 304 0
ix-esc-ca2-07.ix.netcom.com - - [01/Aug/1995:00:00:09 -0400] "GET /images/launch-
logo.gif HTTP/1.0" 200 1713
uplherc.upl.com - - [01/Aug/1995:00:00:10 -0400] "GET /images/WORLD-logosmall.gif
HTTP/1.0" 304 0
slppp6.intermind.net - - [01/Aug/1995:00:00:10 -0400] "GET /history/skylab/
skylab.html HTTP/1.0" 200 1687
piweba4y.prodigy.com - - [01/Aug/1995:00:00:10 -0400] "GET /images/launchmedium.gif
HTTP/1.0" 200 11853
tampico.usc.edu - - [14/Aug/1995:22:57:13 -0400] "GET /welcome.html HTTP/1.0" 200 790
```

Machine syslog file

```
dhcp-47-129:CS100_1> syslog -w 10
Feb  3 15:18:11 dhcp-47-129 Evernote[1140] <Warning>: -[EDAMAccounting
read:]: unexpected field ID 23 with type 8.  Skipping.
Feb  3 15:18:11 dhcp-47-129 Evernote[1140] <Warning>: -[EDAMUser read:]:
unexpected field ID 17 with type 12.  Skipping.
Feb  3 15:18:11 dhcp-47-129 Evernote[1140] <Warning>: -
[EDAMAuthenticationResult read:]: unexpected field ID 6 with type 11.
Skipping.
Feb  3 15:18:11 dhcp-47-129 Evernote[1140] <Warning>: -
[EDAMAuthenticationResult read:]: unexpected field ID 7 with type 11.
Skipping.
Feb  3 15:18:11 dhcp-47-129 Evernote[1140] <Warning>: -[EDAMAccounting
read:]: unexpected field ID 19 with type 8.  Skipping.
Feb  3 15:18:11 dhcp-47-129 Evernote[1140] <Warning>: -[EDAMAccounting
read:]: unexpected field ID 23 with type 8.  Skipping.
Feb  3 15:18:11 dhcp-47-129 Evernote[1140] <Warning>: -[EDAMUser read:]:
unexpected field ID 17 with type 12.  Skipping.
Feb  3 15:18:11 dhcp-47-129 Evernote[1140] <Warning>: -[EDAMSyncState read:]:
unexpected field ID 5 with type 10.  Skipping.
Feb  3 15:18:49 dhcp-47-129 com.apple.mtmd[47] <Notice>: low priority
thinning needed for volume Macintosh HD (/) with 18.9 <= 20.0 pct free space
```

Big Data - Definition and Concepts

- Big Data is not new!
- Traditionally, “Big Data” = **massive volumes of data** [Volume of data at CERN, NASA, Google, ...]
- Where does the Big Data come from?
 - Everywhere! Web logs, RFID, GPS systems, sensor networks, social networks, Internet-based text documents, Internet search indexes, detail call records, astronomy, atmospheric science, biology, genomics, nuclear physics, biochemical experiments, medical records, scientific research, military surveillance, multimedia archives, ...

Fundamentals of Big Data Analytics

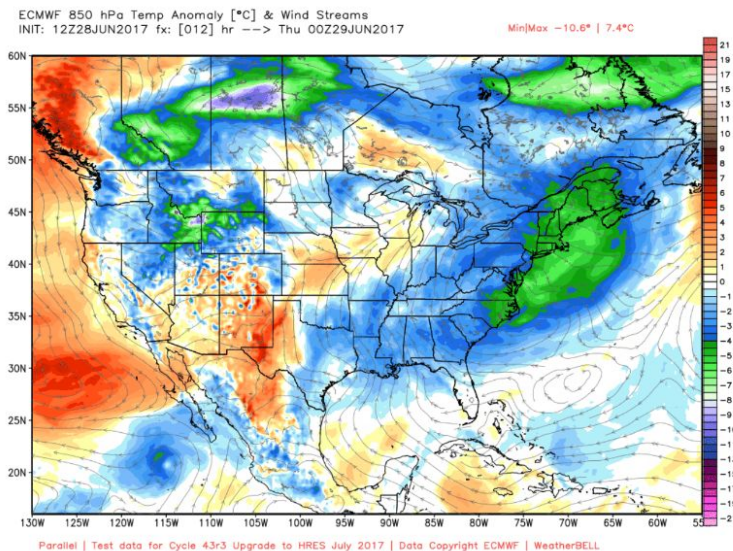
- Big Data by itself, regardless of the size, type, or speed, is worthless
- **Big Data + “big” analytics = value**
- With the value proposition, Big Data also brought about big challenges
 - **Effectively and efficiently capturing, storing, and analyzing** Big Data
 - New breed of technologies needed (developed or purchased or hired or outsourced ...)

Big Data Considerations

- You can't process the amount of data that you want to because of the **limitations** of your current platform.
- You can't include new/contemporary data sources (e.g., social media, RFID, Sensory, Web, GPS, textual data) because it does not comply with the **data storage** schema.
- You need to (or want to) **integrate data as quickly as possible** to be current on your analysis.
- You want to work with a **schema-on-demand data storage** paradigm because of the variety of data types.
- The data is arriving **so fast** at your organization's doorstep that your analytics platform cannot handle it.
- ...

Large-scale computing is not new!

- Weather forecasting has been a long-term scientific Challenge
 - Supercomputers were already used in the 1970s



So, what is the problem?!

- Big data NOW doesn't only concern governments
 - Businesses
 - Individuals
 - Societies
 - Researchers
 - ...
- Main focus is on **how** to make computations on big data **feasible** without a supercomputer
 - Cluster(s) of **commodity hardware**

Challenges of Big Data Analytics

Computational requirement is just a small part of the list of challenges. Following is a list of challenges :

- **Data volume** : The ability to capture, store, and process the huge volume of data in a timely manner
- **Data integration** : The ability to combine data quickly/cost effectively
- **Processing capabilities** :The ability to process the data quickly, as it is captured (i.e., stream analytics)
- **Data governance** (... security, privacy, access) :Skill availability (... data scientist)
- **Solution cost** ()

Critical Success Factors for Big Data Analytics

- A clear **business need** (alignment with the vision and the strategy)
- Strong, committed sponsorship (executive champion)
- Alignment between the business and IT strategy
- A **fact**-based decision-making culture
- A strong data infrastructure
- The **right analytics tools**
- Right people with right **skills**



Enablers of Big Data Analytics

The **need for more efficient analytical systems is increasing**. In order to keep up with the computational needs of Big Data, a **number of new and innovative computational techniques and platforms have been developed**. These techniques are collectively called **high-performance computing**, which includes the following “

- In-memory analytics
 - Storing and processing the complete data set in RAM
- In-database analytics
 - Placing **analytic procedures** close to where **data is stored**
- Grid computing & MPP
 - Use of many machines and processors in **parallel** (MPP - **massively parallel processing**)
- Appliances
 - Combining **hardware, software, and storage in a single unit** for performance and scalability

Business Problems Addressed by Big Data Analytics

- **Process efficiency and cost reduction**
- Brand management
- Revenue maximization, cross-selling/up-selling
- **Enhanced customer experience**
- Churn identification, customer recruiting
- Improved customer service
- Identifying new products and market opportunities
- Risk management
- Regulatory compliance
- Enhanced security capabilities
- ...

Parallelization Challenges

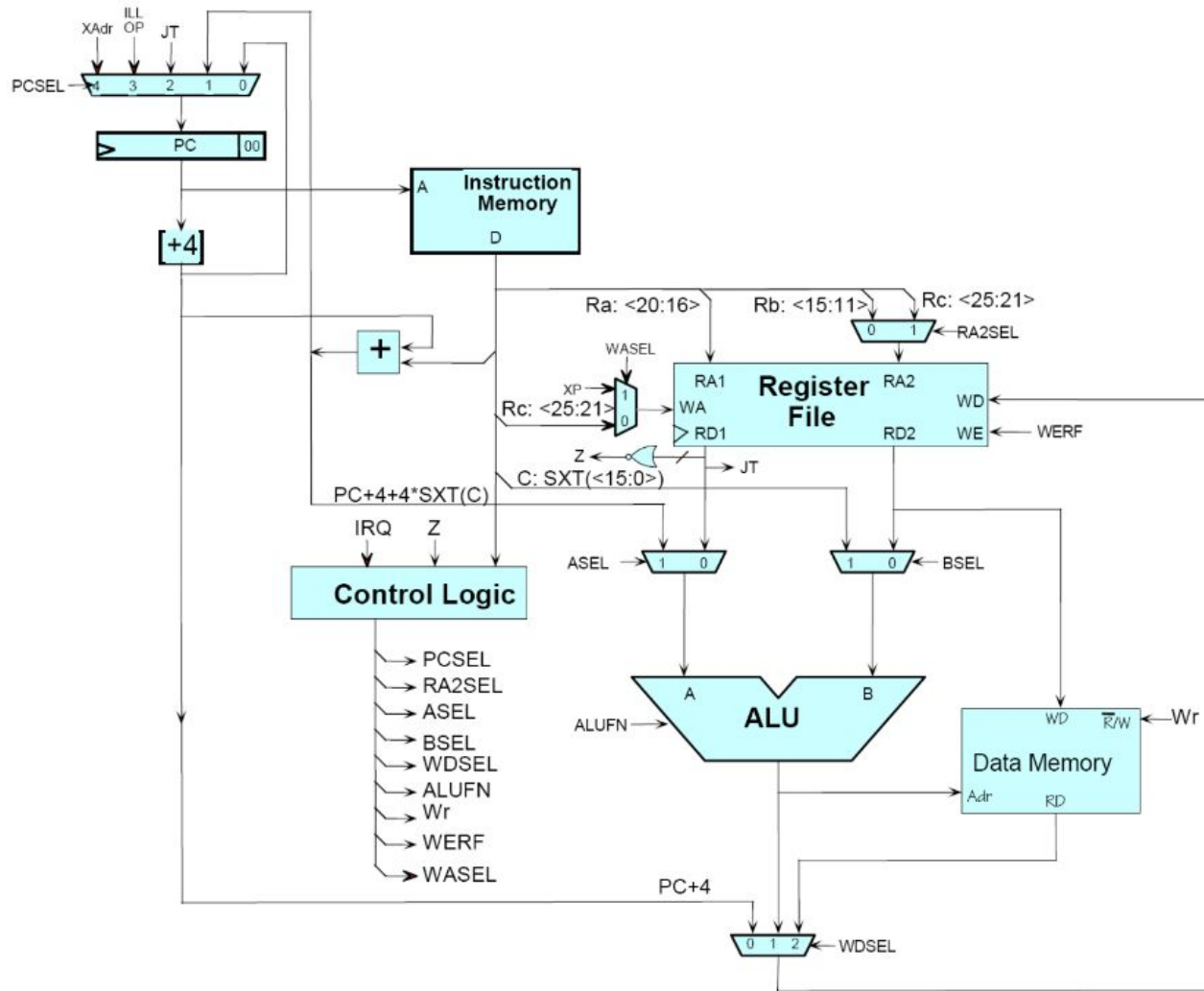
- How do we assign work units to workers?
- What if we have more work units than workers?
- What if some workers are slower than others?
- What if workers need to share partial results?
- How do we aggregate partial results?
- How do we know all the workers have finished?
- What if workers die?

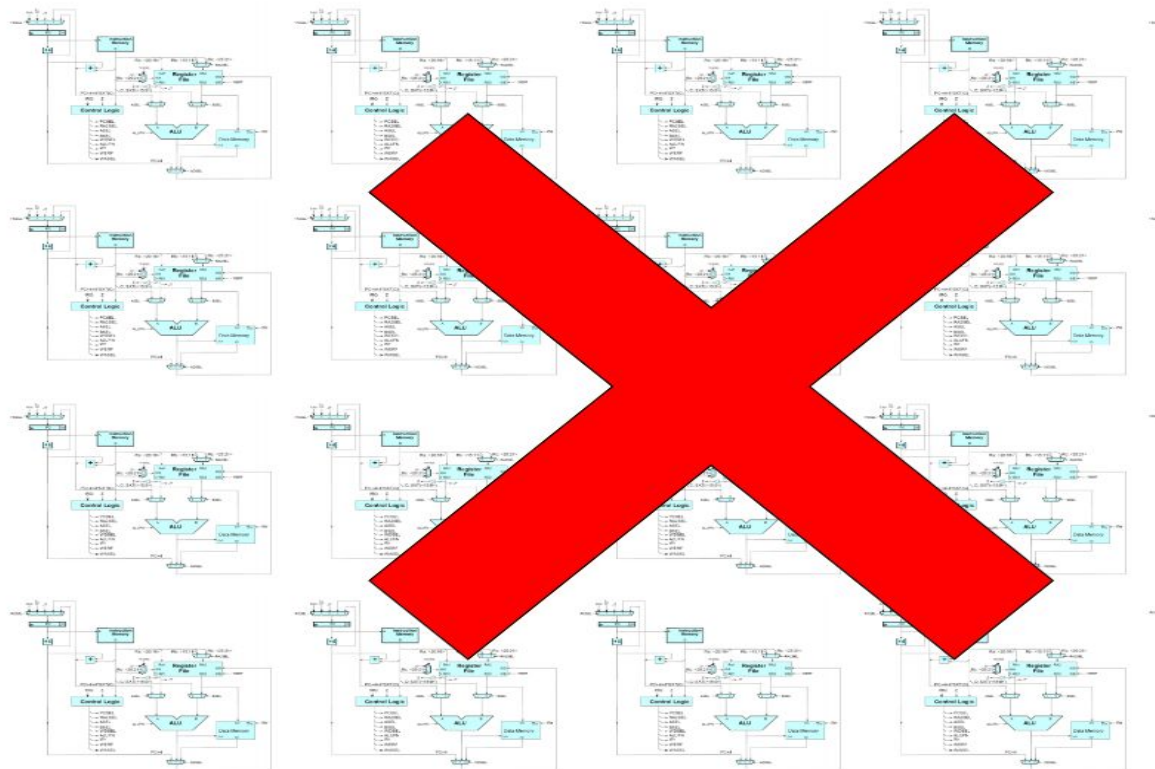
What's the common theme of all of these problems?

- Parallelization problems arise from:
 - **Communication between workers** (e.g., to exchange state)
 - **Access to shared resources** (e.g., data)
- Thus, we need a synchronization mechanism



It's just ***DIFFICULT!***

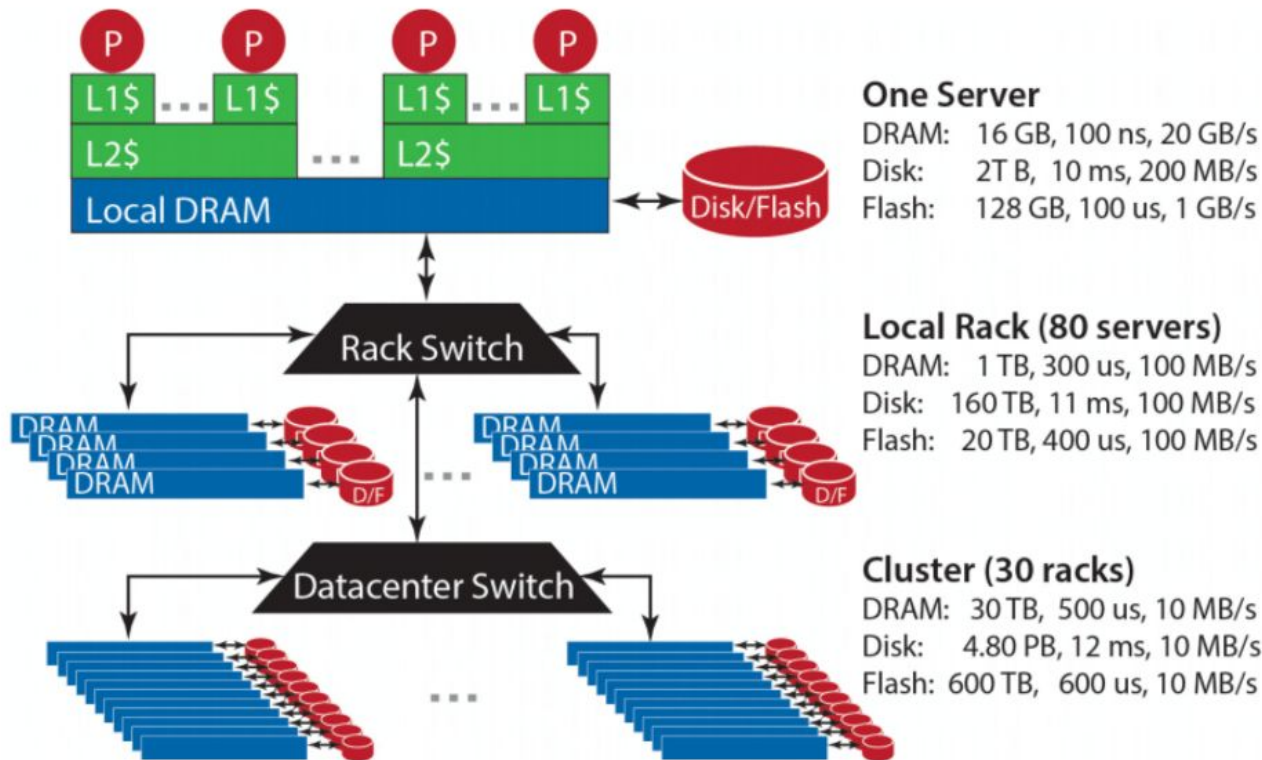


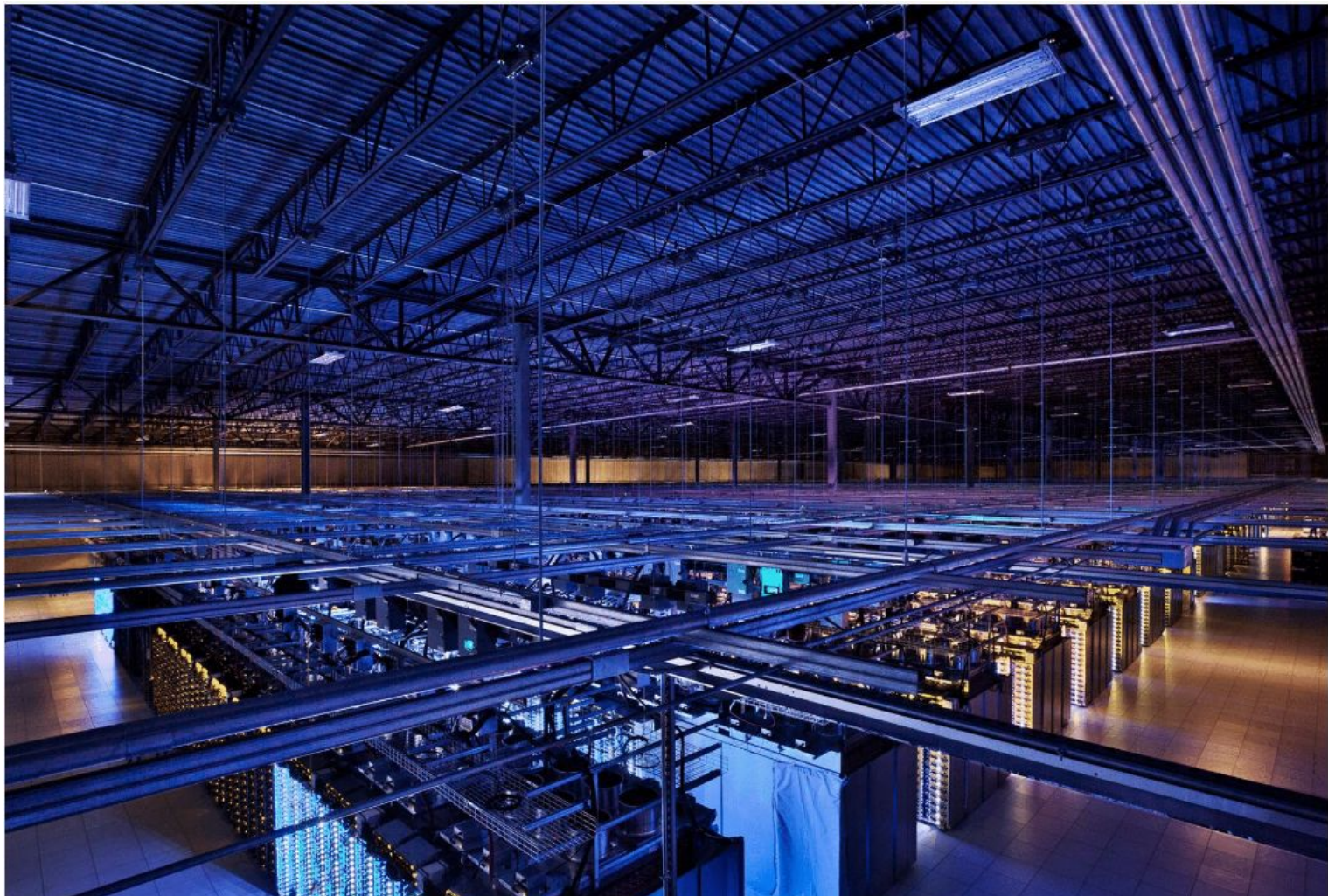


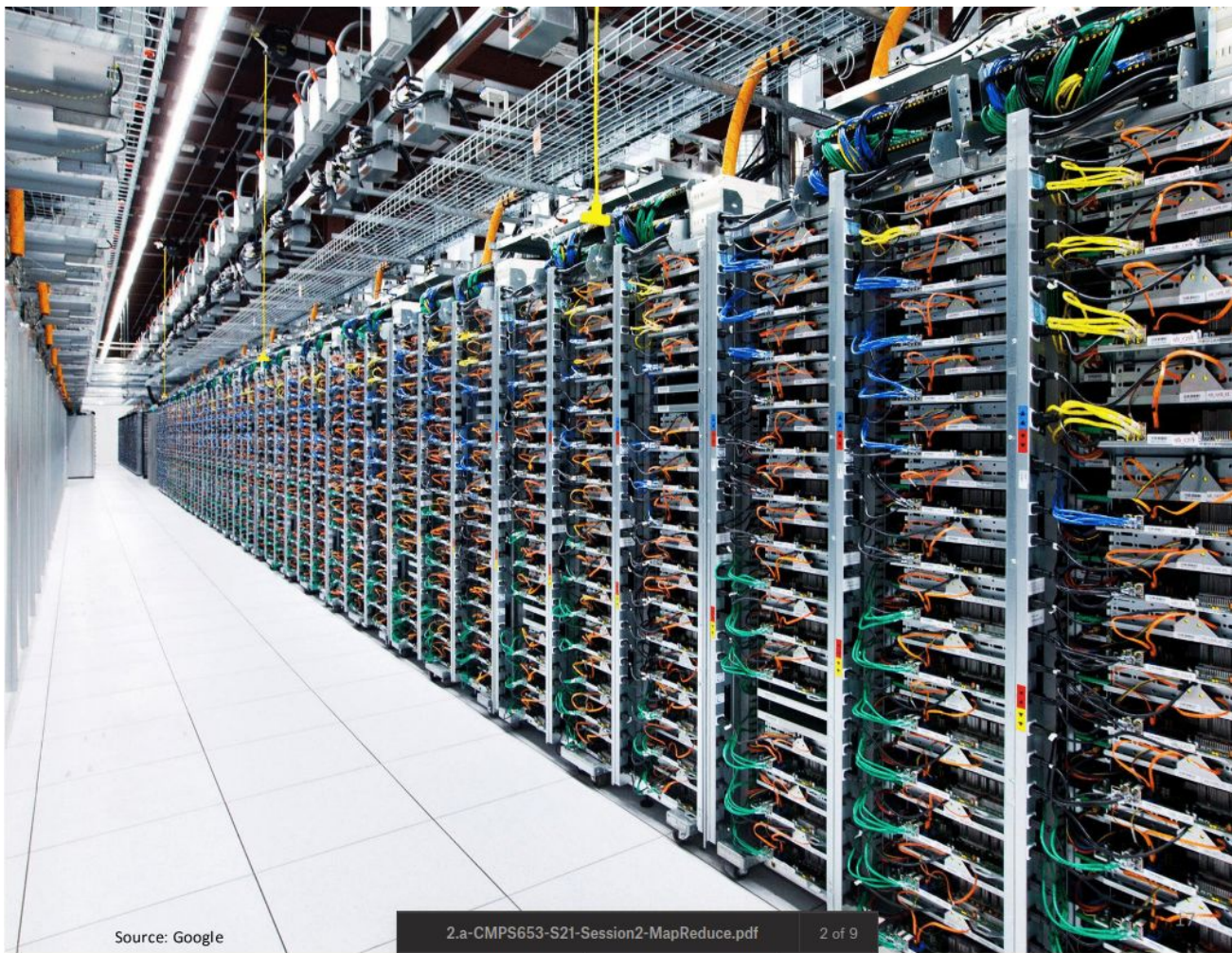
How Data stored



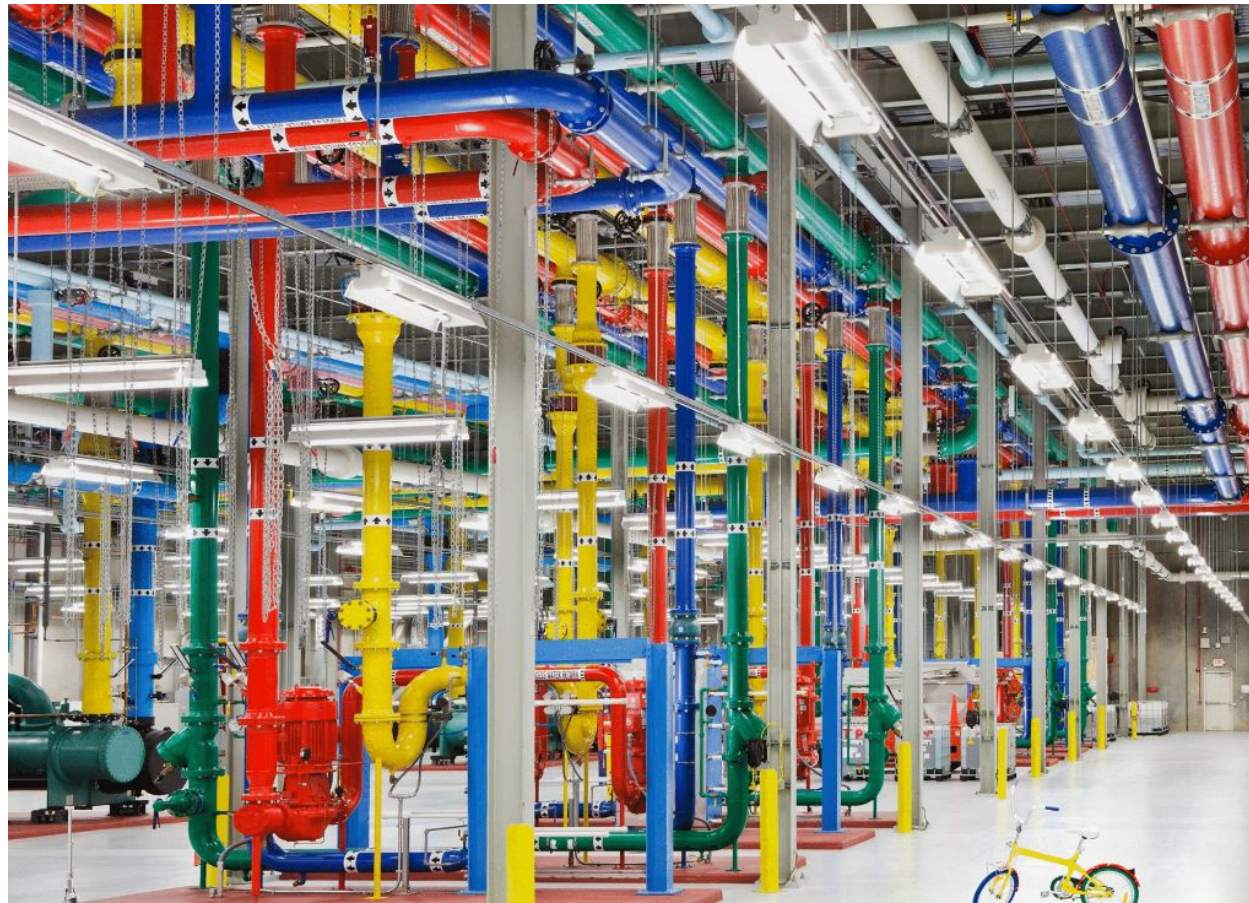
Storage Hierarchy





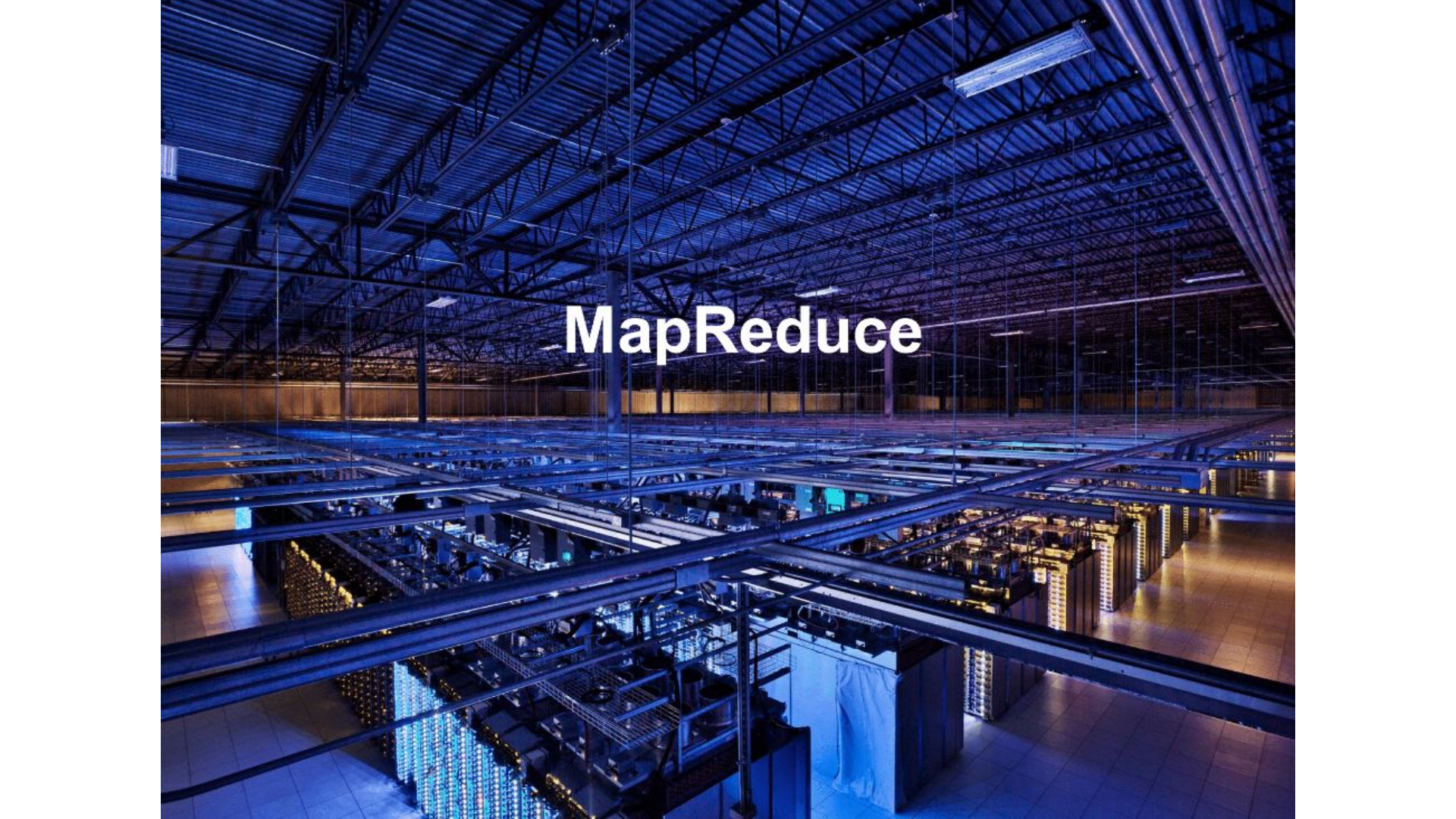


Source: Google



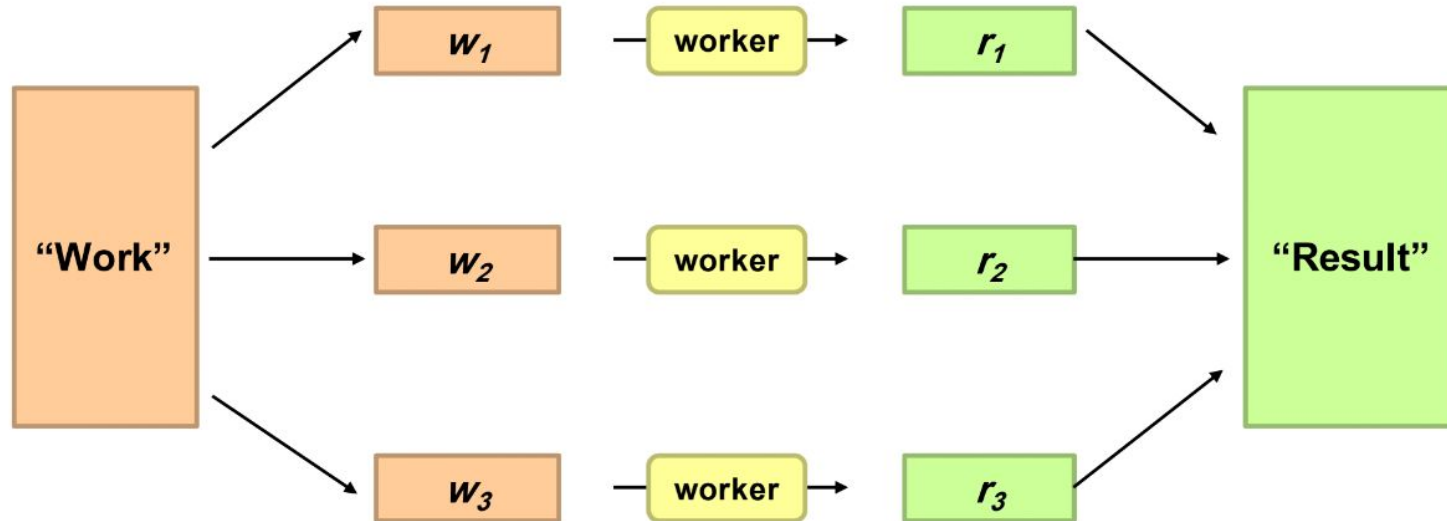


MapReduce

A wide-angle photograph of a massive server room, likely a data center. The room is filled with rows of server racks, some of which are illuminated with blue light. The ceiling is high and features a complex network of metal beams and pipes. The overall atmosphere is industrial and technological.

MapReduce

Divide and Conquer



Parallelization Challenges?

What is MapReduce

A distributed programming framework
for processing

large amount of data

over a cluster of machines

“Big Ideas”

- Scale “out”, not “up”
- Move processing to the data
- Process data sequentially, avoid random access

Seeks vs. Scans

- Consider a 1 TB database with 100 byte records
 - We want to update only 1% of the records (10^8)
- Scenario 1: random access
 - Each update takes ~30 ms (seek, read, write)
 - 10^8 updates = ~35 days
- Scenario 2: rewrite all records
 - Assume 100 MB/s throughput
 - Time = 5.6 hours(!)
- Lesson: avoid random seeks!

The Index scan retrieves all rows from the specified table. T

The Index Seek refers only to the qualified rows and pages

Typical Large-Data Problem

- Iterate over a large number of records

map

- Extract something of interest from each

- Group intermediate results

reduce

- Aggregate intermediate results

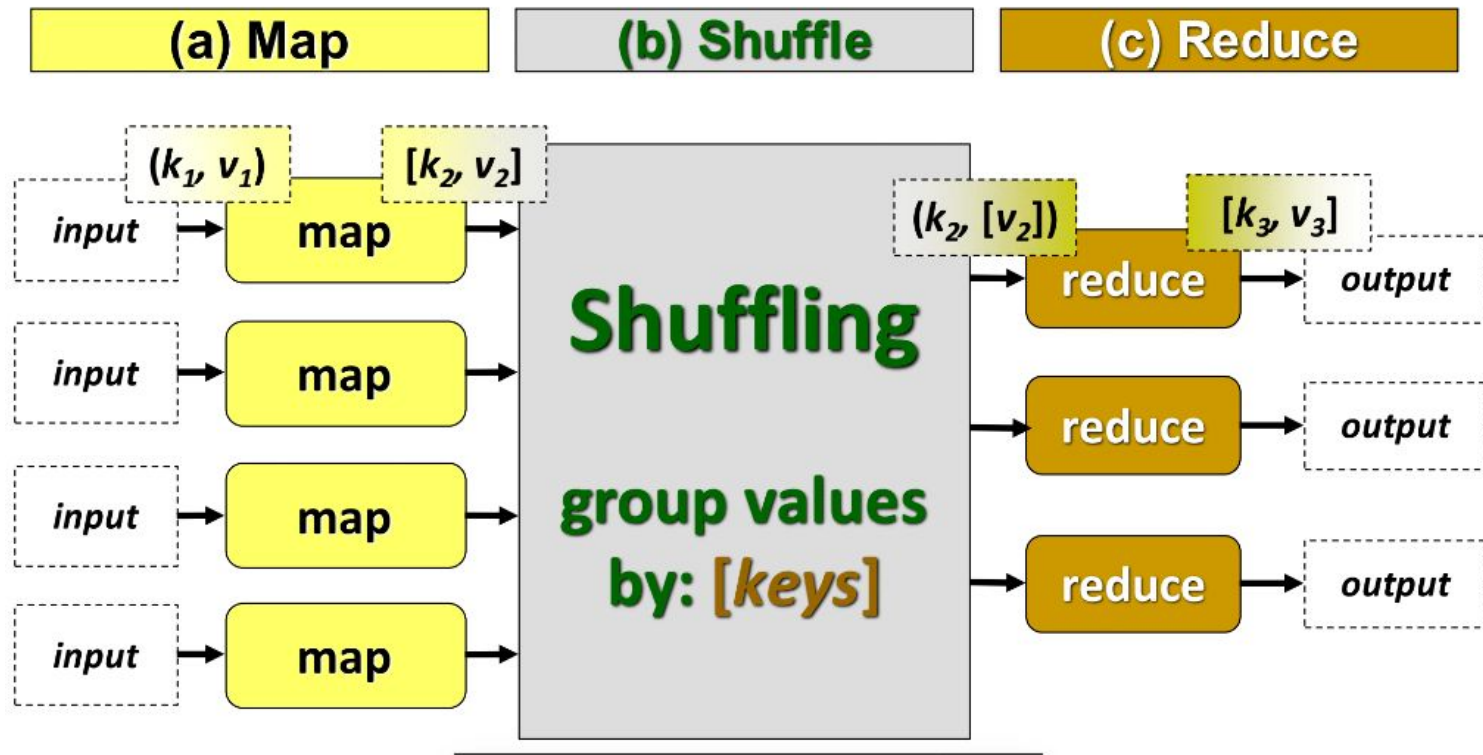
- Write final output

Big Data Technologies MapReduce

- MapReduce **distributes** the processing of very large multi-structured data files across a large cluster of ordinary machines/processors
- Goal - achieving **high performance** with “simple” computers
- Developed and popularized by **Google**
- Good at processing and analyzing large volumes of multi-structured data in a timely manner
- Example tasks: **indexing the Web for search, graph analysis, text analysis, machine learning, ...**

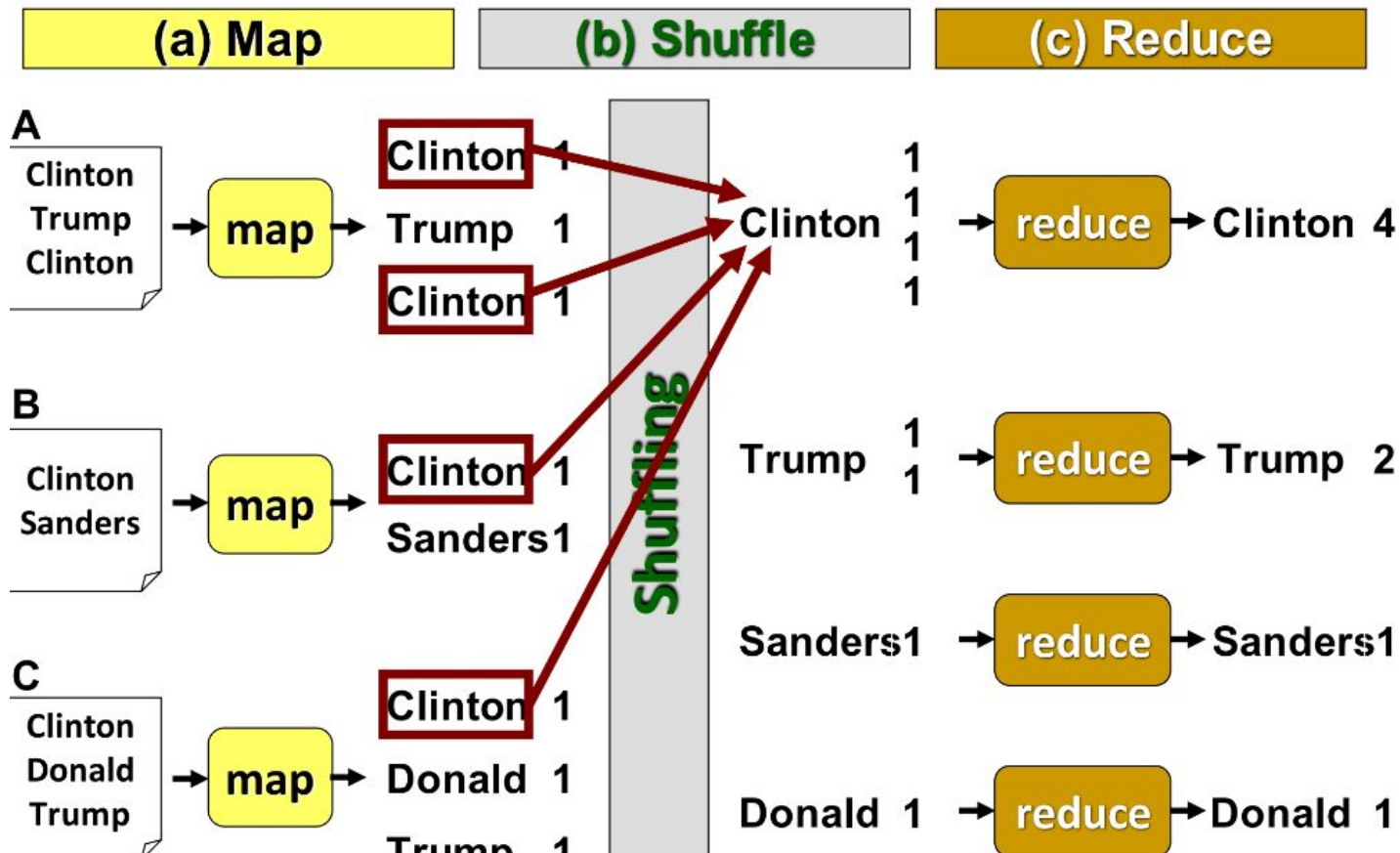


MapReduce Framework

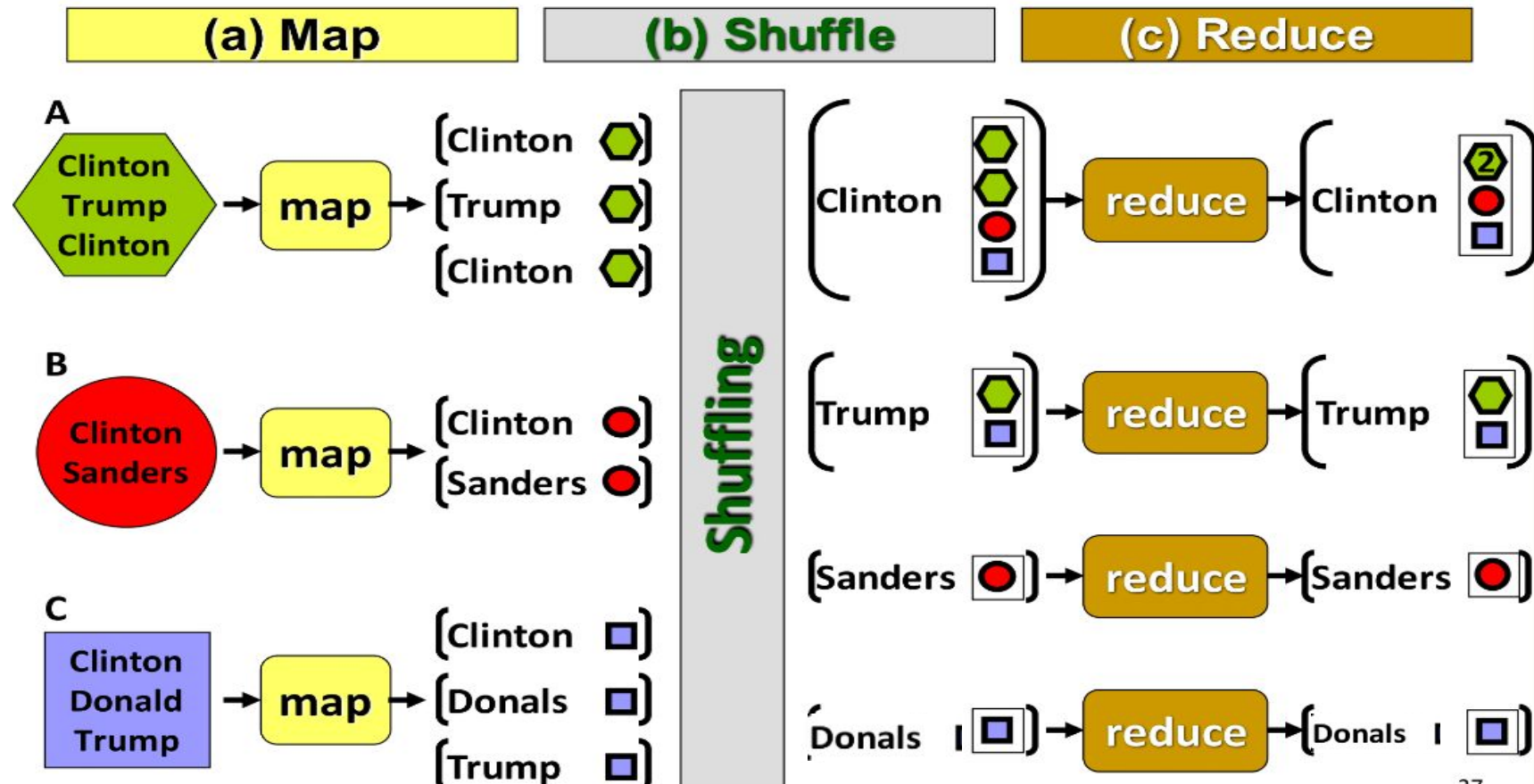


Example: Word Count

Can we compute number of docs per word?



Indexing Example



Big Data Technologies

--Hadoop (1 of 3)

MapReduce Implementations

- Google has a proprietary implementation in C++
- Hadoop is an open-source implementation in Java
 - Development and led by Yahoo, now an Apache project
 - Used in production at Yahoo, Facebook, Twitter, LinkedIn, Netflix, ...
 - The *de facto* big data processing platform
 - Large and expanding software ecosystem



- Lots of custom research implementations
 - For GPUs, cell processors, etc.

Big Data Technologies

--Hadoop (2 of 3)



● How Does Hadoop Work?

- **Access unstructured and semi-structured data** (e.g., log files, social media feeds, other data sources)
- Break the data up into “parts,” which are then loaded into a **file system** made up of multiple nodes running on commodity hardware using **H D F S**
- Each “part” is replicated multiple times and loaded into the file system for replication and failsafe processing
- A node acts as the **Facilitator** and another as **Job Tracker**
- Jobs are distributed to the clients, and once completed the **results are collected and aggregated using MapReduce**

Big Data Technologies

--Hadoop (3 of 3)



- Hadoop Technical Components

- Hadoop Distributed File System (H D F S)
- Name Node (primary facilitator)
- Secondary Node (backup to Name Node)
- Job Tracker
- Slave Nodes (the grunts of any Hadoop cluster)
- Additionally, Hadoop ecosystem is made up of a number of complementary sub-projects: **NoS Q L (Cassandra, Hbase), D W (Hive), ...**
 - NoS Q L = not only S Q L

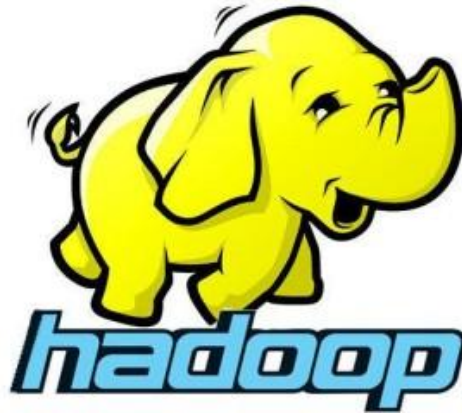
Technology Insights 9.2

A Few Demystifying Facts about Hadoop

- Hadoop consists of multiple products
- Hadoop is open source but available from vendors, too
- Hadoop is an ecosystem, not a single product
- H D F S is a file system, not a D B M S
- Hive resembles S Q L but is not standard S Q L
- Hadoop and MapReduce are related but not the same
- MapReduce provides control for analytics, not analytics
(**MapReduce enables the *infrastructure* for analytics but doesn't provide the analytics themselves.**)

Big Data Technologies

- MapReduce ...
- Hadoop ...
- Hive
- Pig
- Hbase
- Flume
- Oozie
- Ambari
- Avro
- Mahout, Sqoop, Hcatalog,



Big Data Technologies

Hadoop - Demystifying Facts

- Hadoop consists of multiple products
- Hadoop is open source but available from vendors too
- Hadoop is an ecosystem, not a single product
- **HDFS is a file system, not a DBMS**
- Hive resembles SQL but is not standard SQL
- **Hadoop and MapReduce are related but not the same**
- MapReduce provides control for analytics, not analytics
- Hadoop is about data diversity, not just data volume
- Hadoop complements a DW; it's rarely a replacement
- Hadoop **enables many types of analytics**, not just Web analytics

Big Data And Data Warehousing

- What is the impact of Big Data on DW?
 - Big Data and RDBMS do not go nicely together
 - Will Hadoop replace data warehousing/RDBMS?
- Use Cases for Hadoop
 - Hadoop as the repository and refinery
 - Hadoop as the active archive
- Use Cases for Data Warehousing
 - Data warehouse performance
 - Integrating data that provides business value
 - Interactive BI tools

Hadoop versus Data Warehouse

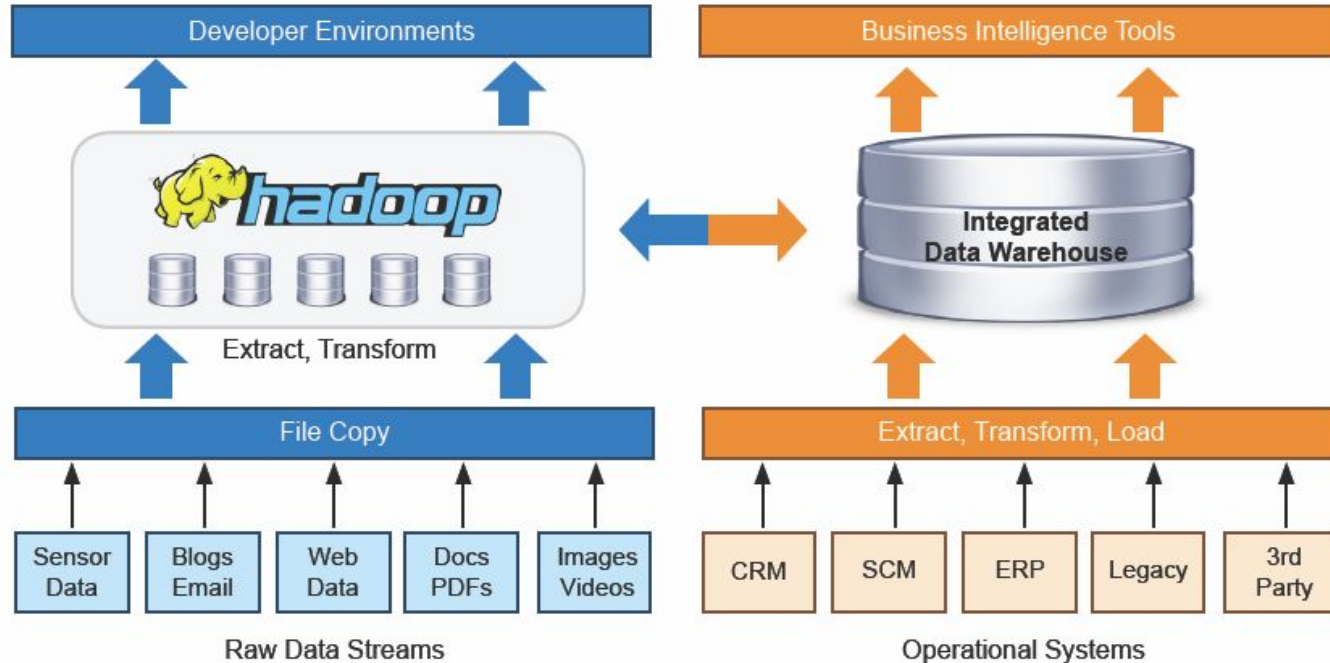
When to Use Which Platform

Requirement	Data Warehouse	Hadoop
Low latency, interactive reports, and OLAP	☑	
ANSI 2003 SQL compliance is required	☑	☑
Preprocessing or exploration of raw unstructured data		☑
Online archives alternative to tape	☑	
High-quality cleansed and consistent data	☑	☑
100s to 1,000s of concurrent users	☑	☑
Discover unknown relationships in the data		☑
Parallel complex process logic	☑	☑
CPU intense analysis	☑	
System, users, and data governance		☑
Many flexible programming languages running in parallel		☑
Unrestricted, ungoverned sandbox explorations		☑
Analysis of provisional data	☑	
Extensive security and regulatory compliance	☑	☑

Coexistence of Hadoop and DW

1. Use Hadoop for storing and archiving multi-structured data
2. Use Hadoop for **filtering, transforming, and/or consolidating multi-structured data**
3. Use Hadoop to **analyze large volumes of multi-structured data and publish the analytical results**
4. Use a relational DBMS that provides MapReduce capabilities as an investigative computing platform
5. Use a **front-end query tool to access and analyze data**

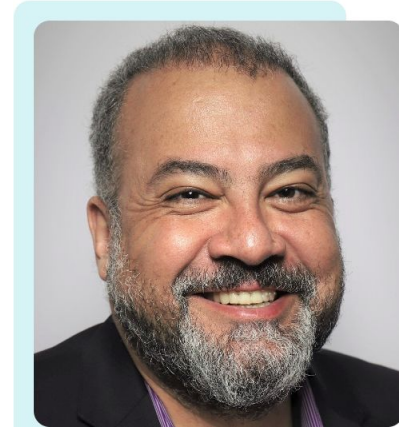
Coexistence of Hadoop and DW



Source: Teradata

Big Data Vendors

- Big Data vendor landscape is developing very rapidly
- A representative list would include
 - Cloudera - cloudera.com
 - MapR – mapr.com
 - Hortonworks - hortonworks.com
 - Also, IBM (Netezza, InfoSphere), Oracle (Exalogic), Microsoft, Amazon, Google, ...



Amr Awadallah

Software,
Hardware,
Service, ...

End of the Chapter

- Questions, comments